

Research Methods Coursework (Part B – Using Generative AI Tools)

Topic: Smart Pharmacy Locator and Drug Availability Management System

1. Research Problem

Access to medication remains a challenge for many patients because they often do not know which nearby pharmacy stocks the medicine they need. In many cases, patients visit several pharmacies before locating the required drug, resulting in wasted time, increased transportation costs, and delayed treatment. Existing map applications can identify pharmacy locations but generally do not provide real-time information on drug availability, medication prices, pharmacist consultation services, or navigation support. As healthcare increasingly embraces digital technologies, there is a growing need for systems that improve access to pharmaceutical services. Therefore, this study proposes a Smart Pharmacy Locator and Drug Availability Management System that enables users to locate nearby pharmacies, check medicine availability and prices, communicate directly with pharmacists, and obtain navigation guidance to selected pharmacies.

2. Review of Related Literature

Recent studies have explored the use of mobile and web technologies to improve access to pharmaceutical services. Akhigbe et al. (2022) developed a mobile-based pharmacy location-aware system that assists users in locating pharmacies where prescribed drugs are available. The system demonstrated good performance and usability on Android devices. However, its focus was primarily on location awareness and did not emphasize pharmacist-patient communication or advanced navigation features.

Wondmagegn et al. (2023) proposed a framework for locating prescribed medication at pharmacies. Their study highlighted the difficulties patients face when searching for medicines and demonstrated how digital solutions can reduce the time spent locating drugs. While the framework improved medication discovery, limited attention was given to real-time pricing information and direct communication with pharmacists.

Irvine (2023) examined the concept of digital medication inventory visibility within healthcare systems. The study emphasized that limited visibility of medicine inventories can negatively affect healthcare delivery and decision-making. The findings suggest that providing real-time inventory information can significantly improve medication accessibility. Nevertheless, the research focused on institutional healthcare settings rather than community pharmacies.

Tinam-isan et al. (2024) introduced the Geospatial Pharmacy Navigator, a web and mobile application integrating Geographic Information Systems (GIS) to identify pharmacies with available prescription drugs. The study demonstrated the usefulness of combining GIS

technology with pharmacy information systems. However, the proposed solution mainly addressed pharmacy discovery and drug availability, with limited support for pharmacist consultation and price comparison.

Kommey et al. (2024) developed MeDASH, a mobile application that uses QR-code technology to provide information on the availability of prescribed drugs in nearby pharmacies. The study showed that digital technologies can improve access to medication information and streamline communication between healthcare providers and pharmacies. Despite these benefits, the system focused mainly on prescription tracking and drug availability rather than comprehensive pharmacy services.

Furthermore, Almeman (2024) discussed the broader digital transformation occurring in the pharmacy sector, emphasizing the growing adoption of online platforms and digital healthcare solutions. The study highlighted the potential of integrated systems that combine information access, communication, and service delivery within pharmaceutical care.

Research Gap

The reviewed studies demonstrate significant progress in pharmacy information systems. However, most existing solutions focus on only one or two aspects of the problem, such as pharmacy location, medicine availability, or inventory visibility. Few systems provide an integrated platform that combines pharmacy discovery, real-time drug availability, medication price comparison, pharmacist-patient communication, and route navigation within a single application. This gap presents an opportunity for the development of a comprehensive smart pharmacy management platform.

3. Proposed Solution and Classification

This study proposes the development of a Smart Pharmacy Locator and Drug Availability Management System. The system will utilize GPS and GIS technologies to identify nearby pharmacies based on the user's current location. Users will be able to search for medicines, view real-time availability information, compare prices across pharmacies, and communicate directly with pharmacists through an integrated messaging feature. Additionally, the application will generate navigation routes to guide users from their current location to the selected pharmacy.

The proposed system aims to reduce the time required to locate medication, improve access to pharmaceutical information, and enhance communication between patients and pharmacists. By integrating multiple services into a single platform, the system seeks to provide a more convenient and efficient healthcare experience.

According to Gregor and Hevner's (2013) Design Science Research Knowledge Contribution Framework, this study is classified as an **Improvement**. Existing pharmacy locator and medication search systems already exist; however, the proposed solution enhances them by

integrating additional functionalities such as real-time inventory visibility, price comparison, pharmacist communication, and navigation support. Rather than creating an entirely new concept, the research improves existing approaches and provides greater value to users through enhanced functionality and usability.

References (APA Style)

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